Type MS · Sector problems with an unknown radius or angle

Answer 1 of 7 · Question 3 part whole · 8 marks

2021 NOV · Paper 01 · extract 1 of 3 · find part whole in this mark scheme extract

3	Mark parts (i) and (ii) together	
	$\frac{r^2\theta}{2}=16.8\Rightarrow\left(\theta=\frac{33.6}{r^2}\right)$	B1
	$r\theta+2r=16.4$	B1
	$r\left(\frac{33.6}{r^2}\right)+2r=16.4\Rightarrow r^2-8.2r+16.8=0$	M1M1
	$r=\frac{8.2\pm\sqrt{(8.2)^2-4\times1\times16.8}}{2\times1}=4,\ (4.2)$	dM1A1
	$\theta=\frac{33.6}{'4'^2}=2.1$	dM1A1 [8]
	ALT	
	$r\left(\frac{33.6}{r^2}\right)+2r=16.4\Rightarrow 5r^2-41r+84=0\Rightarrow(5r-21)(r-4)=0$	[M1M1
	$\Rightarrow r=4,\ \left(\frac{21}{5}\right)$	dM1A1
	$\theta=\frac{33.6}{'4'^2}=2.1$	dM1A1]
	ALT– elimination of r by substitution	
	$\frac{r^2\theta}{2}=16.8$	[B1
	$r\theta+2r=16.4\quad\left(\Rightarrow\quad r=\frac{16.4}{(\theta+2)}\right)$	B1
	$\frac{\left(\frac{16.4}{(\theta+2)}\right)^2}{2}\theta=16.8$	M1
	$33.6\theta^2-134.56\theta+134.4=0$	M1
	$\theta=2.1,(1.90\ldots)$	dM1A1
	$r=\sqrt{\frac{33.6}{'2.1'}}$ or $r=\sqrt{\frac{33.6}{'1.90\ldots'}}\Rightarrow r=4,\left(\frac{21}{5}\right)$	dM1A1
	ALT	B1
	$\frac{r^2\theta}{2}=16.8\Rightarrow\left(r=\sqrt{\frac{33.6}{\theta}}\right)$	B1
	$r\theta+2r=16.4$	M1
	$16.4=2\sqrt{\frac{33.6}{\theta}}+\theta\sqrt{\frac{33.6}{\theta}}\Rightarrow 16.4=(2+\theta)\sqrt{\frac{33.6}{\theta}}$	M1
	$33.6\theta^2-134.56\theta+134.4=0$	dM1A1]

Type MS · Sector problems with an unknown radius or angle

Answer 1 of 7 · Question 3 part whole · 8 marks

2021 NOV · Paper 01 · extract 2 of 3 · find part whole in this mark scheme extract

	$\theta = 2.1, (1.90\ldots)$ $r = \sqrt{\frac{33.6}{\text{'2.1'}}}$ or $r = \sqrt{\frac{33.6}{\text{'1.90...'}}} \Rightarrow r = 4, \left(\frac{21}{5}\right)$	dM1A1
		Total 8 marks

Type MS · Sector problems with an unknown radius or angle

Answer 1 of 7 · Question 3 part whole · 8 marks

2021 NOV · Paper 01 · extract 3 of 3 · find part whole in this mark scheme extract

Question number	Scheme	Marks

Type MS · Sector problems with an unknown radius or angle

Answer 2 of 7 · Question 3 part whole · 6 marks

2023 JUN · Paper 02 · extract 1 of 4 · find part whole in this mark scheme extract

3	**Mark parts (i) and (ii) together.** (Let the obtuse angle $AOB = \alpha$ - students do not need to state this, allow use of any symbol or letter, including θ) $53.2 = r\alpha$ $372.4 = \frac{r^2\alpha}{2}$ $\alpha = \frac{53.2}{r}$, $372.4 = \frac{r^2}{2} \times \frac{53.2}{r}$ oe $\Rightarrow r = 14$ $\alpha = \frac{53.2}{14} = 3.8$ $\theta = (2\pi - 3.8 =) 2.48$	B1 B1 M1A1 M1 A1 [6]
ALT1 final 4 marks	First 2 marks as ALT2 $r = \frac{53.2}{\alpha}, 372.4 = \frac{\left(\frac{53.2}{\alpha}\right)^2}{2} \times \alpha$ oe $\Rightarrow \alpha = \frac{372.4}{1415.12} = 3.8$ $r\left(= \frac{53.2}{3.8}\right) = 14$ $\theta = (2\pi - 3.8 =) 2.48$	B1B1 M1 M1(A1 on ePen) A1(M1 on ePen) A1 [6]
ALT2	$2\pi r - r\theta = 53.2$ oe $\pi r^2 - \frac{1}{2} r^2 \theta = 372.4$ oe $\theta = 2\pi - \frac{53.2}{r} \Rightarrow \pi r^2 - \frac{1}{2}\left(2\pi - \frac{53.2}{r}\right) r^2 = 372.4$ oe $26.6r = 372.4$ oe $r = 14$ $\theta = 2.48$	B1 B1 M1 M1(A1 on ePen) A1(M1 on ePen) A1 [6]

Type MS · Sector problems with an unknown radius or angle

Answer 2 of 7 · Question 3 part whole · 6 marks

2023 JUN · Paper 02 · extract 2 of 4 · find part whole in this mark scheme extract

ALT3	First 2 marks as ALT2	B1B1
	$r = \dfrac{53.2}{2\pi - \theta}$	M1
	$r^2\left(\pi - \dfrac{\theta}{2}\right) = 372.4 \Rightarrow \left(\dfrac{53.2}{2\pi - \theta}\right)^2 = \dfrac{372.4}{\pi - \dfrac{\theta}{2}}$ oe eg $\left(\dfrac{53.2}{2\pi - \theta}\right)^2 = \dfrac{744.8}{2\pi - \theta}$	M1(A1 on ePen)
	$\dfrac{53.2^2}{744.8} = 2\pi - \theta \left(\Rightarrow \theta = 2\pi - \dfrac{53.2^2}{744.8}\right)$	A1(M1 on ePen)
	$\theta = 2.48$	A1
	$r =$ awrt 14.0	[6]
		Total 6 marks

Type MS · Sector problems with an unknown radius or angle

Answer 2 of 7 · Question 3 part whole · 6 marks

2023 JUN · Paper 02 · extract 3 of 4 · find part whole in this mark scheme extract

Mark	Notes
B1	Correctly uses the formula for length of an arc $53.2 = r\times$ angle in radians or correctly uses the formula for length of an arc $53.2 = 2\pi r \times \frac{\text{angle in degrees}}{360}$
B1	Correctly uses the formula for area of a sector $372.4 = \frac{r^2}{2}\times$ angle in radians or correctly uses the formula for area of a sector $372.4 = \pi r^2 \times \frac{\text{angle in degrees}}{360}$
M1	Eliminates α from both equations, allow up to one error. If they have worked in degrees, they **must** reach an equation where π must not be present (ie it has cancelled).
A1	For $r = 14$
M1	For using their r in a correct equation to find a value for α If they are working in degrees, there **must** follow a correct attempt to convert their angle into radians.
A1	For the correct value of $\theta = 2.48$
ALT1 B1B1	As main scheme
M1	Eliminates r from both equations and reaches an equation in α, allow one error
M1 (A1 on ePen)	Solves their equation (allow one error) to find a value for α
A1(M1 on ePen)	For the correct value for r
A1	For the correct value of $\theta = 2.48$
ALT2 B1	Correctly uses the formula for length of an arc
B1	Correctly uses the formula for area of a sector
M1	Rearranges for θ (allow one error) and substitutes correctly into the other equation to reach an equation for r
M1 (A1 on ePen)	For an equation of the form $ar = b$ $ab > 0$
A1(M1 on ePen)	r = 14
A1	$\theta = 2.48$
ALT3 B1B1	As ALT2
M1	Rearranges for r (allow one error) and substitutes correctly into the other equation to reach an equation for θ
M1 (A1 on ePen)	For reaching an equation of the form $d = 2\pi - \theta$ $d > 0$
A1(M1 on ePen)	$\theta = 2.48$
A1	$r =$ awrt 14.0

Type MS · Sector problems with an unknown radius or angle

Answer 2 of 7 · Question 3 part whole · 6 marks

2023 JUN · Paper 02 · extract 4 of 4 · find part whole in this mark scheme extract

Question	Scheme	Marks

Type MS · Sector problems with an unknown radius or angle

Answer 3 of 7 · Question 4 part a · 3 marks

2024 JUN · Paper 01 (regional R) · extract 1 of 15 · find part a in this mark scheme extract

Question	Scheme	Marks
4a)	$\theta = \frac{21-r}{r}$ $A = \frac{r^2}{2}\theta \Rightarrow A = \frac{r^2}{2}\left(\frac{21-r}{r}\right) \Rightarrow A = \frac{r}{2}(21-r)*$ **ALT** OR $\theta r = 21-r$ or $l = 21-r$ $A = \frac{r}{2}r\theta$ or $A = \frac{1}{2}rl \Rightarrow A = \frac{r}{2}(21-r)*$	B1 M1A1 cso [3] [B1 M1A1]
(b)	$27 \leqslant \frac{r}{2}(21-r) \Rightarrow 54 \leqslant 21r - r^2 \Rightarrow r^2 - 21r + 54 \leqslant 0$ Finds c.v's $r^2 - 21r + 54 = 0 \Rightarrow (r-18)(r-3) = 0 \Rightarrow r = 18, 3$ $3 \leqslant r \leqslant 18$	B1 M1 M1A1 [4]
(c)	$\theta = \frac{21-r}{r} \Rightarrow \theta = \frac{21-18}{18} = \frac{1}{6}$ $\theta = \frac{21-3}{3} = 6$ $\Rightarrow \frac{1}{6} \leqslant \theta \leqslant 6$	B1ft B1 [2]
		Total 9 marks

Type MS · Sector problems with an unknown radius or angle

Answer 3 of 7 · Question 4 part a · 3 marks

2024 JUN · Paper 01 (regional R) · extract 2 of 15 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	**Mark parts (i) and (ii) together.** **If they find *a* first score M1M1M1-**	
(i)	M1	For using the correct formula to write down: $4k+1=a+(6-1)d$
	M1	For using the correct formula to write down: $36k+1=\frac{10}{2}(2a+9d)$
	M1	For an explicit, complete method to solve their two simultaneous equations to find a value for d in terms of k, and a value for a Allow **one** processing error. **Note: This is a show question**
(ii)	A1	For the correct expression for d
	A1 cso	For finding $a=-8$ With no errors, this is a given value. Award this if they find only a but not d
(b)	M1	Uses a correct nth term with their expression for d to form a linear equation **and** attempts to solve it.
	A1 cso	For $k=4$ With no errors, this is a given value.
(c)	M1	For setting up the equation as required using the given a and their d No simplification is required for this mark.
	dM1	For forming a 3TQ using the given a and their d Allow one error: $5n^2-71n-K=0$ Or $5n^2-Ln-580=0$ Or $Mn^2-71n-580=0$ Where K, L and M are constants. Note: this mark is dependent on the previous mark being scored.
	M1	For solving their 3TQ using any valid, correct method If they use a calculator and the 3TQ is correct together with the correct final value, award this mark. If the 3TQ is NOT correct, they must show us a correct valid method for solving the 3TQ for the award of this mark.
	A1	For $n=20$ with no other value.

Type MS · Sector problems with an unknown radius or angle

Answer 3 of 7 · Question 4 part a · 3 marks

2024 JUN · Paper 01 (regional R) · extract 3 of 15 · find part a in this mark scheme extract

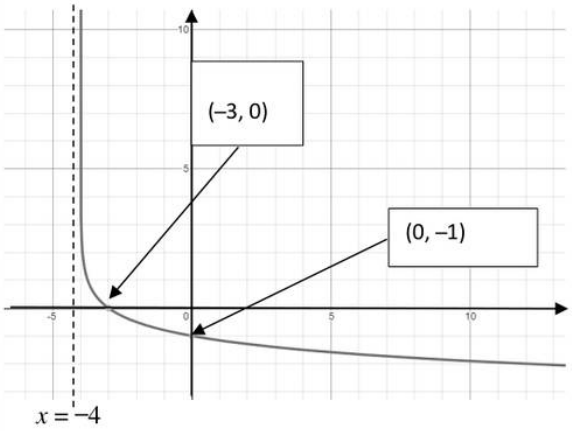
Question	Scheme	Marks
6(a)	$t=0 \Rightarrow s=e^{2\times 0}\sin(3\times 0)+2=2$ $t=\frac{\pi}{6} \Rightarrow s=e^{2\times\frac{\pi}{6}}\sin\left(3\times\frac{\pi}{6}\right)+2=e^{\frac{\pi}{3}}+2$ $AB=e^{\frac{\pi}{3}}+2-2=e^{\frac{\pi}{3}}$ (m)	M1 A1 [2]
(b)	$v=2e^{2t}\sin 3t+3e^{2t}\cos 3t$ When $t=\frac{\pi}{3}$ $v=2e^{\frac{2\pi}{3}}\sin\left(\frac{\pi}{3}\times 3\right)+3e^{\frac{2\pi}{3}}\cos\left(\frac{\pi}{3}\times 3\right)=-3e^{\frac{2\pi}{3}}$ (m/s)	M1A1A1 B1ft [4]
		Total 6 marks

Part	Mark	Notes
(a)	M1	For an attempt to find the displacements of A **and** B as simplified values. At least one must be correct and simplified for this mark and there must be an attempt at the other.
	A1	For finding the distance AB
(b)	M1	For an attempt at product rule. • The formula must be correct. i.e., $[v]=ke^{2t}\times\sin 3t+e^{2t}\times l\cos 3t$ • There must be an acceptable attempt to differentiate both e^{2t} and $\sin 3t$ $e^{2t}\to ke^{2t}$ and $\sin 3t\to l\cos 3t$ $k,l\neq 0$ and integers • The terms must be added.
	A1	One term fully correct.
	A1	Both terms fully correct.
	B1ft	For using $t=\frac{\pi}{3}$ in their v to find a value for v The minimally acceptable expression for v must be a **changed** expression from the given s in terms of e^{2x}, $\sin 3x$ and $\cos 3x$ Note: This is a ft mark. Please check their substitution and the final result. Please isw a value of $-24.36...$ if seen **together** with the exact value.

Type MS · Sector problems with an unknown radius or angle

Answer 3 of 7 · Question 4 part a · 3 marks

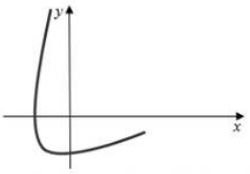
2024 JUN · Paper 01 (regional R) · extract 4 of 15 · find part a in this mark scheme extract

Question	Scheme	Marks
7(a)		B1 Shape and position B1 Intersection with x-axis B1 Intersection with y-axis B1 Asymptote [4]
(b)	$\log_{(x+4)} 256 - \log_4(x+4) = 0 \Rightarrow \dfrac{\log_4 256}{\log_4(x+4)} - \log_4(x+4) = 0$ $\Rightarrow 4 - (\log_4(x+4))^2 = 0 \Rightarrow (\log_4(x+4))^2 = 4$ $\Rightarrow \log_4(x+4) = \pm 2$ $x+4 = 4^2 \Rightarrow x = 12$ $x+4 = 4^{-2} \Rightarrow x = -\dfrac{63}{16}$ or -3.9375	M1 M1 M1 A1 A1 [5]
		Total 9 marks

Type MS · Sector problems with an unknown radius or angle

Answer 3 of 7 · Question 4 part a · 3 marks

2024 JUN · Paper 01 (regional R) · extract 5 of 15 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	B1	For the correct shape in the correct position. Ignore intersections and the asymptote for this mark. Award as long as the curve is the correct way around in quadrants 2, 3 and 4. The ends must not turn back in themselves. Do Not accept for example:  If there are two or more lines, and candidates have not made clear which ONE they want marked– award B0
	B1	For a line/curve **passing** through $(-3, 0)$ or accept $x = -3$ Do not accept stopping at the axis.
	B1	For a line/curve passing through $(0, -1)$ or accept $y = -1$ Do not accept stopping at the axis.
	B1	For the correct **equation** only of the asymptote. If they also give a horizontal asymptote then do not isw – it is B0
(b)	**Method 1 - Works in log base 4**	
	M1	For changing the base of the log to base 4
	M1	For forming a quadratic in terms of $(x + 4)$ Accept a substitution for the log
	M1	For taking the square root **and** undoing the log. Accept just the positive root for this mark.
	A1	For either $x = 12$ or $-\frac{63}{16}$
	A1	For both $x = 12$ and $-\frac{63}{16}$
	Method 2 – Works in log base $(x + 4)$	
	M1	For changing the base of the log to base $(x + 4)$ $\log_{(x+4)} 256 - \frac{\log_{(x+4)}(x+4)}{\log_{(x+4)} 4} = 0 \Rightarrow$
	M1	Changes $\log_{(x+4)}(x+4) = 1$ and $\log_{(x+4)} 256 = 4\log_{(x+4)} 4$ And forming a quadratic equation with log base (x + 4) $\left(\log_{(x+4)} 4\right)^2 = \frac{1}{4}$ Accept a substitution for the log
	M1	Takes the square root (accept just the positive root) **and** undoes the log $\log_{(x+4)} 4 = \pm\frac{1}{2} \Rightarrow 4 = (x+4)^{\frac{1}{2}}$ and $4 = (x+4)^{-\frac{1}{2}}$
	A1	For either $x = 12$ or $-\frac{63}{16}$
	A1	For both $x = 12$ and $-\frac{63}{16}$

Type MS · Sector problems with an unknown radius or angle

Answer 3 of 7 · Question 4 part a · 3 marks

2024 JUN · Paper 01 (regional R) · extract 6 of 15 · find part a in this mark scheme extract

	Method 3 works in an unspecified base. If it is just log, assume base 10	
	M1	For changing the base $\frac{\log 256}{\log(x+4)} - \frac{\log(x+4)}{\log 4} = 0$
	M1	Forms a QE in terms of $(x+4)$ $\quad 4(\log 4)^2 = (\log(x+4))^2$ oe
	M1	Takes square root (accept just the positive root) and undoes the log $\pm 2\log 4 = \log(x+4) \Rightarrow \log 4 = \pm\frac{1}{2}\log(x+4)$ $\Rightarrow \log 4 = \log(x+4)^{\pm\frac{1}{2}} \Rightarrow 4 = (x+4)^{\pm\frac{1}{2}}$ $\Rightarrow 16 = x+4, \quad \frac{1}{16} = x+4$
	A1	For either $x = 12$ or $-\frac{63}{16}$
	,A1	For both correct values $x = 12$ and $-\frac{63}{16}$

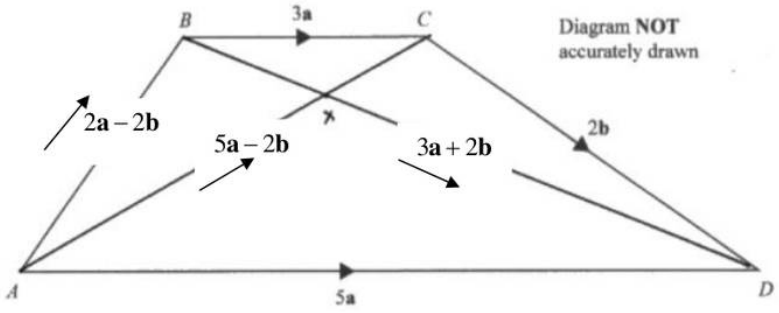
Type MS · Sector problems with an unknown radius or angle

Answer 3 of 7 · Question 4 part a · 3 marks

2024 JUN · Paper 01 (regional R) · extract 7 of 15 · find part a in this mark scheme extract

Question	Scheme	Marks
8(a)	$\overrightarrow{AB} = \overrightarrow{AD} + \overrightarrow{DC} + \overrightarrow{CB} = 5\mathbf{a} - 2\mathbf{b} - 3\mathbf{a} = 2\mathbf{a} - 2\mathbf{b}$	B1 [1]
(b)	$\overrightarrow{BX} = k\overrightarrow{BD} = k(3\mathbf{a} + 2\mathbf{b})$ $\overrightarrow{BX} = \overrightarrow{BA} + \lambda\overrightarrow{AC} = -2\mathbf{a} + 2\mathbf{b} + \lambda(5\mathbf{a} - 2\mathbf{b})$ $\Rightarrow k(3\mathbf{a} + 2\mathbf{b}) = -2\mathbf{a} + 2\mathbf{b} + \lambda(5\mathbf{a} - 2\mathbf{b})$ $\Rightarrow 3k\mathbf{a} + 2k\mathbf{b} = (5\lambda - 2)\mathbf{a} + (2 - 2\lambda)\mathbf{b}$ $\Rightarrow 3k = 5\lambda - 2 \quad 2k = 2 - 2\lambda$ $\Rightarrow k = \frac{3}{8} \quad \left[\lambda = \frac{5}{8}\right]$	M1 M1 M1 M1A1 [5]
(c)	$\Delta CXD = \frac{5}{8}\Delta BCD$ $\Delta BCD = \frac{3}{8}ABCD$ $\frac{\Delta CXD}{ABCD} = \frac{\frac{5}{8}}{\frac{3}{8}} = \frac{15}{64}$ Ratio of area of triangle CXD : Trapezium $ABCD$ = 15 : 64	M1 M1 M1 A1 [4]
		Total 10 marks

Useful sketch



Type MS · Sector problems with an unknown radius or angle

Answer 3 of 7 · Question 4 part a · 3 marks

2024 JUN · Paper 01 (regional R) · extract 8 of 15 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	B1	For the correct vector for $\overrightarrow{AB}$
(b)	**Note carefully.** For the first two marks in part (b) you must follow through their working using their $\overrightarrow{AB}$ or $\overrightarrow{AC}$ or $\overrightarrow{BD}$	
	M1	For one correct vector statement for $\overrightarrow{BX}$
	M1	For a second correct and different vector statement for $\overrightarrow{BX}$ For example: $\overrightarrow{BX} = \overrightarrow{BA} + \lambda\overrightarrow{AC} = -2\mathbf{a} + 2\mathbf{b} + \lambda(5\mathbf{a} - 2\mathbf{b})$ OR $\overrightarrow{BX} = \overrightarrow{BC} + \lambda\overrightarrow{AC} = 3\mathbf{a} - \lambda(5\mathbf{a} - 2\mathbf{b})$
	ddM1	For equating coefficients of **a** and **b** and forming two equations in k and another parameter. This must be correct **Dependent on first two M marks**
	dddM1	For solving their simultaneous equations to find k Allow one arithmetical error in processing. **Dependent on all three previous M marks**
	A1	For $k = \frac{3}{8}$
(c)	M1	For using their k to find the ratio of the areas of for example, $CXD : CAD \Rightarrow \frac{CXD}{CAD} = \frac{3}{8}$ or $\frac{CXD}{CAD} = \frac{\frac{1}{2}\times\frac{3}{8}\times 2b\times\sin ACD}{\frac{1}{2}\times\frac{8}{8}\times 2b\times\sin ACD} = \frac{3}{8}$ **OR** $CXD : BCD \Rightarrow \frac{CXD}{BCD} = \frac{5}{8}$ or $\frac{CXD}{CAD} = \frac{\frac{1}{2}\times\frac{5}{8}\times 2b\times\sin BDC}{\frac{1}{2}\times\frac{8}{8}\times 2b\times\sin BDC} = \frac{5}{8}$
	M1	For using the given lengths of BC and AD [3 and 5],to find the ratio of the area of triangle ACD : trapezium $ABCD$ $\frac{CAD}{ABCD} = \frac{\frac{1}{2}\times h\times 5}{\frac{1}{2}\times h\times(5+3)} = \frac{5}{8}$ **OR** $\frac{BCD}{ABCD} = \frac{\frac{1}{2}\times h\times 3}{\frac{1}{2}\times h\times(5+3)} = \frac{3}{8}$
	Note carefully. You must ft their ratios above and check to see if they are being combined correctly for the final M mark here.	
	M1	Combines the areas above to obtain: $\frac{CXD}{CAD}\times\frac{CAD}{ABCD} = \frac{3}{8}\times\frac{5}{8} = \left(\frac{15}{64}\right)$ **OR** $\frac{CXD}{BCD}\times\frac{BCD}{ABCD} = \frac{5}{8}\times\frac{3}{8} = \left(\frac{15}{64}\right)$
	A1	For the correct ratio 15 : 64

Type MS · Sector problems with an unknown radius or angle

Answer 3 of 7 · Question 4 part a · 3 marks

2024 JUN · Paper 01 (regional R) · extract 9 of 15 · find part a in this mark scheme extract

Question	Scheme	Marks
9(a)	$\frac{y-8}{8-3}=\frac{x-6}{6--4}\Rightarrow\frac{y-8}{5}=\frac{x-6}{10}\Rightarrow\left[y-8=\frac{1}{2}(x-6)\right]$	M1A1 [2]
(b)	$C=\left(\frac{4\times6+1\times-4}{4+1},\frac{4\times8+1\times3}{4+1}\right)=(4,7)$	B1,B1 [2]
(c)	$-2=\frac{q-7}{p-4}\Rightarrow q=15-2p$ $8\sqrt{5}=\sqrt{(4-p)^2+(7-q)^2}$ $\Rightarrow 320=(4-p)^2+(2p-8)^2\Rightarrow p^2-8p-48=0$ $\Rightarrow(p+4)(p-12)=0$ $\Rightarrow p=-4\quad q=15-2\times-4=23\Rightarrow D=(-4,23)$	M1 M1 M1 M1 A1A1 [6]
(d)	Area of triangle $ACD=\frac{1}{2}\times(23-3)\times(4--4)=80$ [square units] **ALT 1** $\frac{1}{2}\begin{bmatrix}-4&4&-4&-4\\3&7&23&3\end{bmatrix}=\frac{1}{2}\left[(-28+92-12)-(12-28-92)\right]=80$ **ALT 2** $A=\frac{1}{2}\times AC\times CD=\frac{1}{2}\times4\sqrt{5}\times8\sqrt{5}=80$	M1A1 [2] [M1A1] [M1A1]
		Total 12 marks

Part	Mark	Notes
(a)	M1	For a **complete** method to find an equation of the line segment AB They must either use the formula shown Award M1A1 for $\frac{y-8}{5}=\frac{x-6}{10}$ (the denominators must be processed). OR They must find the gradient using a correct method and then use the formula $y-y_1=m(x-x_1)\Rightarrow y-8='\frac{1}{2}'(x-6)$ or using $(-4,3)$ OR They must find the gradient using a correct method and then using $y=mx+c$ they must find c and **put an equation together** to score this mark. $y='\frac{1}{2}'x+'5'$ Allow for example $k=\frac{1}{2}x+5$ for this mark (M1) only.
	A1	For a correct equation in any form.[See above]

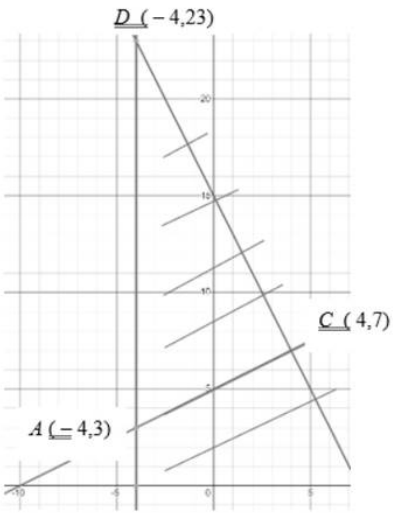
Type MS · Sector problems with an unknown radius or angle

Answer 3 of 7 · Question 4 part a · 3 marks

2024 JUN · Paper 01 (regional R) · extract 10 of 15 · find part a in this mark scheme extract

(b)	B1	For either correct coordinate of C (4, 7) **This is an M mark in Epen**
	B1	For both correct coordinates of C (4, 7) **This is an M mark in Epen**
(c)	**Note:** You must ft their coords of C in part (c)	
	M1	For using the perpendicular gradient of their equation in (a) to set up a correct equation in terms of p and q **OR** For finding the equation of the line CD using their point C (4, 7) The method must be complete for this mark. $\frac{y-7}{7-23}=\frac{x-4}{4--4} \Rightarrow y=-2x+15$
	M1	For using the given length of CD to set up a second equation using Pythagoras theorem in terms of p and q $8\sqrt{5}=\sqrt{(4-p)^2+(7-q)^2}$ oe
	M1	For combining their two equations in p and q to obtain a 3TQ
	M1	For solving their 3TQ (provided it is a 3TQ) by any acceptable method. If the 3TQ is incorrect, there must be evidence of a method to score this mark. If they use their calculator on a correct 3TQ and obtain the correct values, score M1.
	A1	For the correct value of p or q
	A1	For the correct values of both p and q
(d)	M1	For any correct method to find the area of triangle ACD, using their coordinates. If they use the determinant method, check their work carefully.
	A1	For the correct area of 80 [square units].

Useful sketch



Type MS · Sector problems with an unknown radius or angle

Answer 3 of 7 · Question 4 part a · 3 marks

2024 JUN · Paper 01 (regional R) · extract 11 of 15 · find part a in this mark scheme extract

Question	Scheme	Marks
10(a)	$\alpha+\beta=-\frac{k}{2}\qquad \alpha\beta=2$	B1
	$(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta,\quad \alpha^2-\beta^2=(\alpha+\beta)(\alpha-\beta)$	B1B1
	$\Rightarrow \alpha^2-\beta^2=(\alpha+\beta)\sqrt{(\alpha+\beta)^2-4\alpha\beta}$	
	$\Rightarrow \alpha^2-\beta^2=\left(-\frac{k}{2}\right)\sqrt{\left(-\frac{k}{2}\right)^2-4\times 2}=\frac{7\sqrt{17}}{4}$	M1
	$\Rightarrow \frac{k^4}{4}-8k^2-\frac{833}{4}=0\Rightarrow k^4-32k^2-833=0$ or any equivalent	M1A1
	$\Rightarrow (k^2-49)(k^2+17)=0\Rightarrow k=-7$ *	M1A1 cso [8]
(b)	Product: $\left[(\alpha-\beta)(\alpha+\beta)=\alpha^2-\beta^2=\frac{7\sqrt{17}}{4}\right]$ from (a)	B1
	Sum: $(\alpha-\beta)+(\alpha+\beta)=\frac{\sqrt{17}}{2}+\frac{7}{2}=\left(\frac{\sqrt{17}+7}{2}\right)$	M1
	$x^2-\left(\frac{\sqrt{17}+7}{2}\right)x+\frac{7\sqrt{17}}{4}=0\Rightarrow 4x^2-2\left(\sqrt{17}+7\right)x+7\sqrt{17}=0$	M1A1 [4]
		Total 12 marks

There is an alternative solution using the roots of the equation.

Roots are $\frac{-k\pm\sqrt{k^2-4\times 2\times 4}}{2\times 2}$

$$\Rightarrow\left(\frac{-k+\sqrt{k^2-32}}{4}\right)^2-\left(\frac{-k-\sqrt{k^2-32}}{4}\right)^2=\frac{7\sqrt{17}}{4}$$

$$\Rightarrow\left(-k+\sqrt{k^2-32}\right)^2-\left(-k-\sqrt{k^2-32}\right)^2=28\sqrt{17}$$

$$\Rightarrow k^2-2k\sqrt{k^2-32}+k^2-32-k^2-2k\sqrt{k^2-32}-k^2+32=28\sqrt{17}$$

$$\Rightarrow -4k\sqrt{k^2-32}=28\sqrt{17}\Rightarrow k\sqrt{k^2-32}=7\sqrt{17}$$

$$\Rightarrow k^2\left(k^2-32\right)=833\Rightarrow k^4-32k^2-833=0$$

$$\Rightarrow (k^2-49)(k^2+17)=0\Rightarrow k=-7\ *$$

Type MS · Sector problems with an unknown radius or angle

Answer 3 of 7 · Question 4 part a · 3 marks

2024 JUN · Paper 01 (regional R) · extract 12 of 15 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	B1	For the correct sum in terms of k and the correct product. Look out for these embedded in their work.
	B1	For the correct algebra on $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta$ Substitution not required for this mark, although it can be implied from correct substitution without the actual algebra seen.
	B1	For the correct algebra on $\alpha^2-\beta^2$ Award also for $(\alpha+\beta)(\alpha-\beta)=\frac{7\sqrt{17}}{4}$ Substitution not required for this mark, although it can be implied from correct substitution without the actual algebra seen.
	M1	For substituting in the values of their sum and product into their $\alpha^2-\beta^2$ This can be implied from sight of $\left(-\frac{k}{2}\right)\sqrt{\left(-\frac{k}{2}\right)^2-4\times2}=\frac{7\sqrt{17}}{4}$
	M1	For forming a 3TQ in terms of k^2 in any form
	A1	For the correct 3TQ Accept it in any form as long as it is 3 terms. For example, accept $\frac{k^4}{16}-2k^2-\frac{833}{16}=0$ etc
	M1	For any acceptable attempt seen to solve their 3TQ [see General Guidance] If there is no working, just $k=-7$ following a 3TQ then award M0 However, accept evidence of $k^2=49$ [with $k^2=-17$]
	A1 cso	For the correct value of k * **Note: This is a given value**
(b)	B1	For a value for the product using their results from (a) or just writes it down.
	M1	For a correct method to find the value of the sum using their values for $(\alpha-\beta)$ and $(\alpha+\beta)$
	M1	For forming a 3TQ with their sum and product. = 0 is not required for the award of this mark
	A1	For a correct equation simplified or unsimplified. For example, accept, $x^2-\left(\frac{\sqrt{17}+7}{2}\right)x+\frac{7\sqrt{17}}{4}=0$ as well as $4x^2-2\left(\sqrt{17}+7\right)x+7\sqrt{17}=0$ o.e.

Type MS · Sector problems with an unknown radius or angle

Answer 3 of 7 · Question 4 part a · 3 marks

2024 JUN · Paper 01 (regional R) · extract 13 of 15 · find part a in this mark scheme extract

Question	Scheme	Marks
11(a)	$f(\theta)=(2\cos\theta-\sin\theta)(2\sin\theta+\cos\theta)$ $=4\sin\theta\cos\theta+2\cos^2\theta-2\sin^2\theta-\sin\theta\cos\theta$ $=3\sin\theta\cos\theta+2(\cos^2\theta-\sin^2\theta)\Rightarrow\frac{3}{2}\sin 2\theta,+2\cos 2\theta$ $=\frac{3}{2}\sin 2\theta+2\cos 2\theta$ *	M1 M1A1 cso [3]
(b)	$\frac{3}{2}\sin 2\theta+2\cos 2\theta+2=0\Rightarrow 3\sin 2\theta+4\cos 2\theta+4=0$ $\Rightarrow 3\sin 2\theta+4(\cos^2\theta-\sin^2\theta)+4(\sin^2\theta+\cos^2\theta)=0$ $\Rightarrow 6\sin\theta\cos\theta+8\cos^2\theta=0$ $\Rightarrow\cos\theta(6\sin\theta+8\cos\theta)=0$ $\Rightarrow\cos\theta=0\Rightarrow\theta=\frac{\pi}{2}$ * $\left[\Rightarrow\tan\theta=-\frac{4}{3}\right]$	M1M1 M1 M1 A1 cso [5]
(c)	Area $=\int_0^{\frac{\pi}{2}}\left(\frac{3}{2}\sin 2\theta+2\cos 2\theta+2\right)d\theta=\left[-\frac{3}{4}\cos 2\theta+\frac{2\sin 2\theta}{2}+2\theta\right]_0^{\frac{\pi}{2}}$ $=\left(-\frac{3}{4}\cos 2\left(\frac{\pi}{2}\right)+\frac{2\sin 2\left(\frac{\pi}{2}\right)}{2}+2\left(\frac{\pi}{2}\right)\right)-\left(-\frac{3}{4}\cos 0+\frac{2\sin 0}{2}+2\times 0\right)$ $=\left(\frac{3}{4}+\pi\right)-\left(-\frac{3}{4}\right)=\frac{3}{2}+\pi$	M1 M1 A1 [3]
		Total 11 marks

Part	Mark	Notes
(a)	M1	For multiplying out the brackets **correctly** and simplifying to $k\sin\theta\cos\theta$, $l(\cos^2\theta-\sin^2\theta)$ where k and l are integers. Condone invisible brackets if the working is clear.
	M1	For using the identity for $\cos 2\theta$ on their $2(\cos^2\theta-\sin^2\theta)$ **OR** For using the identity for $\sin 2\theta$ on their $3\sin\theta\cos\theta$
	A1 cso	For the correct identity as shown with no errors. You must check every line of working carefully – this is a given answer.
	Solutions based on RHS = LHS are acceptable. Apply the above marks – if you are not sure, please send to REVIEW	

Type MS · Sector problems with an unknown radius or angle

Answer 3 of 7 · Question 4 part a · 3 marks

2024 JUN · Paper 01 (regional R) · extract 14 of 15 · find part a in this mark scheme extract

(b)	M1	For correctly using the identity $\cos 2\theta = \cos^2\theta - \sin^2\theta$
	M1	For correctly using the identity $\sin^2\theta + \cos^2\theta = 1$
	M1	For correctly using the identity $\sin 2\theta = 2\sin\theta\cos\theta$
	M1	For factorising the resulting expression. You must see this step
	A1 cso	For $\cos\theta = 0 \Rightarrow \theta = \frac{\pi}{2}$
	ALT	
	M1	For correctly using $\cos 2\theta = \cos^2\theta - \sin^2\theta$ $\frac{3}{2}\sin 2\theta + 2\cos 2\theta + 2 = \frac{3}{2}\sin 2\theta + 2\left(\cos^2\theta - \sin^2\theta\right) = 0$
	M1	For correctly using the identity $\sin^2\theta + \cos^2\theta = 1$ $\frac{3}{2}\sin 2\theta + 2\cos 2\theta + 2 = \frac{3}{2}\sin 2\theta + 2\left(\cos^2\theta - 1 + \cos^2\theta\right) + 2 = 0$ $\Rightarrow \left[\frac{3}{2}\sin 2\theta + 4\cos^2\theta = 0\right]$
	M1	For correctly using the identity $\sin 2\theta = 2\sin\theta\cos\theta$ $\frac{3}{2}\sin 2\theta + 4\cos^2\theta = 3\sin\theta\cos\theta + 4\cos^2\theta = 0$
	M1	For factorising the resulting expression. You must see this step $3\sin\theta\cos\theta + 4\cos^2\theta = \cos\theta\left(3\sin\theta + 4\cos\theta\right) = 0$
	A1	$\cos\theta = 0 \Rightarrow \theta = \frac{\pi}{2}$ *
(c)	M1	For integrating the given expression. Minimally acceptable integration as shown below: $\frac{3}{2}\sin 2\theta \Rightarrow \pm\frac{3}{4}\cos 2\theta$ $2\theta\cos 2\theta \Rightarrow \pm\left(\frac{2}{2}\right)\sin 2\theta$ $2 \Rightarrow 2\theta$
	M1	For substituting **BOTH** of the given limits the correct way around into **their changed expression** and subtracting.. There must be a minimum of two terms to substitute limits into. This must be explicitly seen
	A1	For the correct exact area as shown. [Approximate area = 4.641….]

Type MS · Sector problems with an unknown radius or angle

Answer 3 of 7 · Question 4 part a · 3 marks

2024 JUN · Paper 01 (regional R) · extract 15 of 15 · find part a in this mark scheme extract

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Type MS · Sector problems with an unknown radius or angle

Answer 4 of 7 · Question 4 part c · 2 marks

2024 JUN · Paper 01 (regional R) · extract 1 of 15 · find part c in this mark scheme extract

Question	Scheme	Marks
4a)	$\theta = \dfrac{21-r}{r}$ $A = \dfrac{r^2}{2}\theta \Rightarrow A = \dfrac{r^2}{2}\left(\dfrac{21-r}{r}\right) \Rightarrow A = \dfrac{r}{2}(21-r)*$ **ALT** OR $\theta r = 21-r$ or $l = 21-r$ $A = \dfrac{r}{2}r\theta$ or $A = \dfrac{1}{2}rl \Rightarrow A = \dfrac{r}{2}(21-r)*$	B1 M1A1 cso [3] [B1 M1A1]
(b)	$27 \leqslant \dfrac{r}{2}(21-r) \Rightarrow 54 \leqslant 21r - r^2 \Rightarrow r^2 - 21r + 54 \leqslant 0$ Finds c.v's $r^2 - 21r + 54 = 0 \Rightarrow (r-18)(r-3) = 0 \Rightarrow r = 18, 3$ $3 \leqslant r \leqslant 18$	B1 M1 M1A1 [4]
(c)	$\theta = \dfrac{21-r}{r} \Rightarrow \theta = \dfrac{21-18}{18} = \dfrac{1}{6}$ $\theta = \dfrac{21-3}{3} = 6$ $\Rightarrow \dfrac{1}{6} \leqslant \theta \leqslant 6$	B1ft B1 [2]
		Total 9 marks

Type MS · Sector problems with an unknown radius or angle

Answer 4 of 7 · Question 4 part c · 2 marks

2024 JUN · Paper 01 (regional R) · extract 2 of 15 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	**Mark parts (i) and (ii) together.** **If they find *a* first score M1M1M1-**	
	M1	For using the correct formula to write down: $4k+1=a+(6-1)d$
	M1	For using the correct formula to write down: $36k+1=\frac{10}{2}(2a+9d)$
(i)	M1	For an explicit, complete method to solve their two simultaneous equations to find a value for d in terms of k, and a value for a Allow **one** processing error. **Note: This is a show question**
(ii)	A1	For the correct expression for d
	A1 cso	For finding $a=-8$ With no errors, this is a given value. Award this if they find only a but not d
(b)	M1	Uses a correct nth term with their expression for d to form a linear equation **and** attempts to solve it.
	A1 cso	For $k=4$ With no errors, this is a given value.
(c)	M1	For setting up the equation as required using the given a and their d No simplification is required for this mark.
	dM1	For forming a 3TQ using the given a and their d Allow one error: $5n^2-71n-K=0$ Or $5n^2-Ln-580=0$ Or $Mn^2-71n-580=0$ Where K, L and M are constants. Note: this mark is dependent on the previous mark being scored.
	M1	For solving their 3TQ using any valid, correct method If they use a calculator and the 3TQ is correct together with the correct final value, award this mark. If the 3TQ is NOT correct, they must show us a correct valid method for solving the 3TQ for the award of this mark.
	A1	For $n=20$ with no other value.

Type MS · Sector problems with an unknown radius or angle

Answer 4 of 7 · Question 4 part c · 2 marks

2024 JUN · Paper 01 (regional R) · extract 3 of 15 · find part c in this mark scheme extract

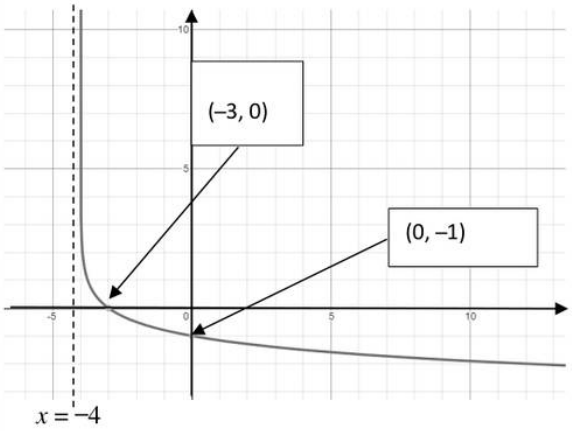
Question	Scheme	Marks
6(a)	$t=0 \Rightarrow s=e^{2\times 0}\sin(3\times 0)+2=2$ $t=\frac{\pi}{6} \Rightarrow s=e^{2\times\frac{\pi}{6}}\sin\left(3\times\frac{\pi}{6}\right)+2=e^{\frac{\pi}{3}}+2$ $AB=e^{\frac{\pi}{3}}+2-2=e^{\frac{\pi}{3}}$ (m)	M1 A1 [2]
(b)	$v=2e^{2t}\sin 3t+3e^{2t}\cos 3t$ When $t=\frac{\pi}{3}$ $v=2e^{\frac{2\pi}{3}}\sin\left(\frac{\pi}{3}\times 3\right)+3e^{\frac{2\pi}{3}}\cos\left(\frac{\pi}{3}\times 3\right)=-3e^{\frac{2\pi}{3}}$ (m/s)	M1A1A1 B1ft [4]
		Total 6 marks

Part	Mark	Notes
(a)	M1	For an attempt to find the displacements of A **and** B as simplified values. At least one must be correct and simplified for this mark and there must be an attempt at the other.
	A1	For finding the distance AB
(b)	M1	For an attempt at product rule. • The formula must be correct. i.e., $[v]=ke^{2t}\times\sin 3t+e^{2t}\times l\cos 3t$ • There must be an acceptable attempt to differentiate both e^{2t} and $\sin 3t$ $e^{2t}\to ke^{2t}$ and $\sin 3t\to l\cos 3t$ $k,l\neq 0$ and integers • The terms must be added.
	A1	One term fully correct.
	A1	Both terms fully correct.
	B1ft	For using $t=\frac{\pi}{3}$ in their v to find a value for v The minimally acceptable expression for v must be a **changed** expression from the given s in terms of e^{2x}, $\sin 3x$ and $\cos 3x$ Note: This is a ft mark. Please check their substitution and the final result. Please isw a value of $-24.36...$ if seen **together** with the exact value.

Type MS · Sector problems with an unknown radius or angle

Answer 4 of 7 · Question 4 part c · 2 marks

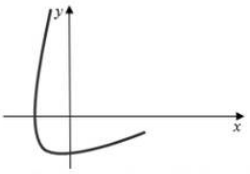
2024 JUN · Paper 01 (regional R) · extract 4 of 15 · find part c in this mark scheme extract

Question	Scheme	Marks
7(a)		B1 Shape and position B1 Intersection with x-axis B1 Intersection with y-axis B1 Asymptote [4]
(b)	$\log_{(x+4)} 256 - \log_4(x+4) = 0 \Rightarrow \dfrac{\log_4 256}{\log_4(x+4)} - \log_4(x+4) = 0$ $\Rightarrow 4 - (\log_4(x+4))^2 = 0 \Rightarrow (\log_4(x+4))^2 = 4$ $\Rightarrow \log_4(x+4) = \pm 2$ $x+4 = 4^2 \Rightarrow x = 12$ $x+4 = 4^{-2} \Rightarrow x = -\dfrac{63}{16}$ or -3.9375	M1 M1 M1 A1 A1 [5]
		Total 9 marks

Type MS · Sector problems with an unknown radius or angle

Answer 4 of 7 · Question 4 part c · 2 marks

2024 JUN · Paper 01 (regional R) · extract 5 of 15 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	B1	For the correct shape in the correct position. Ignore intersections and the asymptote for this mark. Award as long as the curve is the correct way around in quadrants 2, 3 and 4. The ends must not turn back in themselves. Do Not accept for example:  If there are two or more lines, and candidates have not made clear which ONE they want marked– award B0
	B1	For a line/curve **passing** through $(-3, 0)$ or accept $x = -3$ Do not accept stopping at the axis.
	B1	For a line/curve passing through $(0, -1)$ or accept $y = -1$ Do not accept stopping at the axis.
	B1	For the correct **equation** only of the asymptote. If they also give a horizontal asymptote then do not isw – it is B0
(b)	**Method 1 - Works in log base 4**	
	M1	For changing the base of the log to base 4
	M1	For forming a quadratic in terms of $(x + 4)$ Accept a substitution for the log
	M1	For taking the square root **and** undoing the log. Accept just the positive root for this mark.
	A1	For either $x = 12$ or $-\frac{63}{16}$
	A1	For both $x = 12$ and $-\frac{63}{16}$
	Method 2 – Works in log base $(x + 4)$	
	M1	For changing the base of the log to base $(x + 4)$ $\log_{(x+4)} 256 - \frac{\log_{(x+4)}(x+4)}{\log_{(x+4)} 4} = 0 \Rightarrow$
	M1	Changes $\log_{(x+4)}(x+4) = 1$ and $\log_{(x+4)} 256 = 4\log_{(x+4)} 4$ And forming a quadratic equation with log base (x + 4) $\left(\log_{(x+4)} 4\right)^2 = \frac{1}{4}$ Accept a substitution for the log
	M1	Takes the square root (accept just the positive root) **and** undoes the log $\log_{(x+4)} 4 = \pm\frac{1}{2} \Rightarrow 4 = (x+4)^{\frac{1}{2}}$ and $4 = (x+4)^{-\frac{1}{2}}$
	A1	For either $x = 12$ or $-\frac{63}{16}$
	A1	For both $x = 12$ and $-\frac{63}{16}$

Type MS · Sector problems with an unknown radius or angle

Answer 4 of 7 · Question 4 part c · 2 marks

2024 JUN · Paper 01 (regional R) · extract 6 of 15 · find part c in this mark scheme extract

	Method 3 works in an unspecified base. If it is just log, assume base 10	
	M1	For changing the base $\frac{\log 256}{\log(x+4)} - \frac{\log(x+4)}{\log 4} = 0$
	M1	Forms a QE in terms of $(x+4)$ $\quad 4(\log 4)^2 = (\log(x+4))^2$ oe
	M1	Takes square root (accept just the positive root) and undoes the log $\pm 2\log 4 = \log(x+4) \Rightarrow \log 4 = \pm\frac{1}{2}\log(x+4)$ $\Rightarrow \log 4 = \log(x+4)^{\pm\frac{1}{2}} \Rightarrow 4 = (x+4)^{\pm\frac{1}{2}}$ $\Rightarrow 16 = x+4, \quad \frac{1}{16} = x+4$
	A1	For either $x = 12$ or $-\frac{63}{16}$
	,A1	For both correct values $x = 12$ and $-\frac{63}{16}$

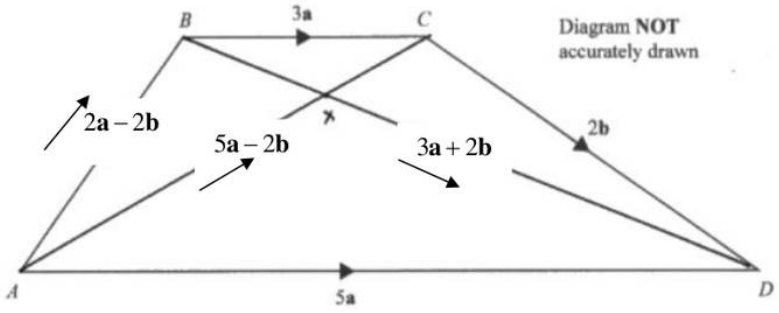
Type MS · Sector problems with an unknown radius or angle

Answer 4 of 7 · Question 4 part c · 2 marks

2024 JUN · Paper 01 (regional R) · extract 7 of 15 · find part c in this mark scheme extract

Question	Scheme	Marks
8(a)	$\overrightarrow{AB} = \overrightarrow{AD} + \overrightarrow{DC} + \overrightarrow{CB} = 5\mathbf{a} - 2\mathbf{b} - 3\mathbf{a} = 2\mathbf{a} - 2\mathbf{b}$	B1 [1]
(b)	$\overrightarrow{BX} = k\overrightarrow{BD} = k(3\mathbf{a} + 2\mathbf{b})$ $\overrightarrow{BX} = \overrightarrow{BA} + \lambda\overrightarrow{AC} = -2\mathbf{a} + 2\mathbf{b} + \lambda(5\mathbf{a} - 2\mathbf{b})$ $\Rightarrow k(3\mathbf{a} + 2\mathbf{b}) = -2\mathbf{a} + 2\mathbf{b} + \lambda(5\mathbf{a} - 2\mathbf{b})$ $\Rightarrow 3k\mathbf{a} + 2k\mathbf{b} = (5\lambda - 2)\mathbf{a} + (2 - 2\lambda)\mathbf{b}$ $\Rightarrow 3k = 5\lambda - 2 \quad 2k = 2 - 2\lambda$ $\Rightarrow k = \frac{3}{8} \quad \left[\lambda = \frac{5}{8}\right]$	M1 M1 M1 M1A1 [5]
(c)	$\Delta CXD = \frac{5}{8}\Delta BCD$ $\Delta BCD = \frac{3}{8}ABCD$ $\frac{\Delta CXD}{ABCD} = \frac{\frac{5}{8}}{\frac{3}{8}} = \frac{15}{64}$ Ratio of area of triangle CXD : Trapezium $ABCD$ = 15 : 64	M1 M1 M1 A1 [4]
		Total 10 marks

Useful sketch



Type MS · Sector problems with an unknown radius or angle

Answer 4 of 7 · Question 4 part c · 2 marks

2024 JUN · Paper 01 (regional R) · extract 8 of 15 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	B1	For the correct vector for $\overrightarrow{AB}$
(b)	**Note carefully.** For the first two marks in part (b) you must follow through their working using their $\overrightarrow{AB}$ or $\overrightarrow{AC}$ or $\overrightarrow{BD}$	
	M1	For one correct vector statement for $\overrightarrow{BX}$
	M1	For a second correct and different vector statement for $\overrightarrow{BX}$ For example: $\overrightarrow{BX} = \overrightarrow{BA} + \lambda\overrightarrow{AC} = -2\mathbf{a} + 2\mathbf{b} + \lambda(5\mathbf{a} - 2\mathbf{b})$ OR $\overrightarrow{BX} = \overrightarrow{BC} + \lambda\overrightarrow{AC} = 3\mathbf{a} - \lambda(5\mathbf{a} - 2\mathbf{b})$
	ddM1	For equating coefficients of **a** and **b** and forming two equations in k and another parameter. This must be correct **Dependent on first two M marks**
	dddM1	For solving their simultaneous equations to find k Allow one arithmetical error in processing. **Dependent on all three previous M marks**
	A1	For $k = \frac{3}{8}$
(c)	M1	For using their k to find the ratio of the areas of for example, $CXD : CAD \Rightarrow \frac{CXD}{CAD} = \frac{3}{8}$ or $\frac{CXD}{CAD} = \frac{\frac{1}{2}\times\frac{3}{8}\times 2b\times\sin ACD}{\frac{1}{2}\times\frac{8}{8}\times 2b\times\sin ACD} = \frac{3}{8}$ **OR** $CXD : BCD \Rightarrow \frac{CXD}{BCD} = \frac{5}{8}$ or $\frac{CXD}{CAD} = \frac{\frac{1}{2}\times\frac{5}{8}\times 2b\times\sin BDC}{\frac{1}{2}\times\frac{8}{8}\times 2b\times\sin BDC} = \frac{5}{8}$
	M1	For using the given lengths of BC and AD [3 and 5],to find the ratio of the area of triangle ACD : trapezium $ABCD$ $\frac{CAD}{ABCD} = \frac{\frac{1}{2}\times h\times 5}{\frac{1}{2}\times h\times(5+3)} = \frac{5}{8}$ **OR** $\frac{BCD}{ABCD} = \frac{\frac{1}{2}\times h\times 3}{\frac{1}{2}\times h\times(5+3)} = \frac{3}{8}$
	Note carefully. You must ft their ratios above and check to see if they are being combined correctly for the final M mark here.	
	M1	Combines the areas above to obtain: $\frac{CXD}{CAD}\times\frac{CAD}{ABCD} = \frac{3}{8}\times\frac{5}{8} = \left(\frac{15}{64}\right)$ **OR** $\frac{CXD}{BCD}\times\frac{BCD}{ABCD} = \frac{5}{8}\times\frac{3}{8} = \left(\frac{15}{64}\right)$
	A1	For the correct ratio 15 : 64

Type MS · Sector problems with an unknown radius or angle

Answer 4 of 7 · Question 4 part c · 2 marks

2024 JUN · Paper 01 (regional R) · extract 9 of 15 · find part c in this mark scheme extract

Question	Scheme	Marks
9(a)	$\frac{y-8}{8-3}=\frac{x-6}{6--4}\Rightarrow\frac{y-8}{5}=\frac{x-6}{10}\Rightarrow\left[y-8=\frac{1}{2}(x-6)\right]$	M1A1 [2]
(b)	$C=\left(\frac{4\times6+1\times-4}{4+1},\frac{4\times8+1\times3}{4+1}\right)=(4,7)$	B1,B1 [2]
(c)	$-2=\frac{q-7}{p-4}\Rightarrow q=15-2p$ $8\sqrt{5}=\sqrt{(4-p)^2+(7-q)^2}$ $\Rightarrow 320=(4-p)^2+(2p-8)^2\Rightarrow p^2-8p-48=0$ $\Rightarrow(p+4)(p-12)=0$ $\Rightarrow p=-4\quad q=15-2\times-4=23\Rightarrow D=(-4,23)$	M1 M1 M1 M1 A1A1 [6]
(d)	Area of triangle $ACD=\frac{1}{2}\times(23-3)\times(4--4)=80$ [square units] **ALT 1** $\frac{1}{2}\begin{bmatrix}-4&4&-4&-4\\3&7&23&3\end{bmatrix}=\frac{1}{2}\left[(-28+92-12)-(12-28-92)\right]=80$ **ALT 2** $A=\frac{1}{2}\times AC\times CD=\frac{1}{2}\times4\sqrt{5}\times8\sqrt{5}=80$	M1A1 [2] [M1A1] [M1A1]
		Total 12 marks

Part	Mark	Notes
(a)	M1	For a **complete** method to find an equation of the line segment AB They must either use the formula shown Award M1A1 for $\frac{y-8}{5}=\frac{x-6}{10}$ (the denominators must be processed). OR They must find the gradient using a correct method and then use the formula $y-y_1=m(x-x_1)\Rightarrow y-8='\frac{1}{2}'(x-6)$ or using $(-4,3)$ OR They must find the gradient using a correct method and then using $y=mx+c$ they must find c and **put an equation together** to score this mark. $y='\frac{1}{2}'x+'5'$ Allow for example $k=\frac{1}{2}x+5$ for this mark (M1) only.
	A1	For a correct equation in any form.[See above]

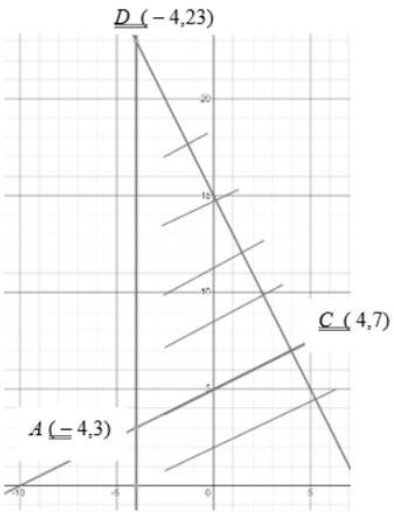
Type MS · Sector problems with an unknown radius or angle

Answer 4 of 7 · Question 4 part c · 2 marks

2024 JUN · Paper 01 (regional R) · extract 10 of 15 · find part c in this mark scheme extract

(b)	B1	For either correct coordinate of C (4, 7) **This is an M mark in Epen**
	B1	For both correct coordinates of C (4, 7) **This is an M mark in Epen**
(c)	**Note:** You must ft their coords of C in part (c)	
	M1	For using the perpendicular gradient of their equation in (a) to set up a correct equation in terms of p and q **OR** For finding the equation of the line CD using their point C (4, 7) The method must be complete for this mark. $\frac{y-7}{7-23}=\frac{x-4}{4--4} \Rightarrow y=-2x+15$
	M1	For using the given length of CD to set up a second equation using Pythagoras theorem in terms of p and q $8\sqrt{5}=\sqrt{(4-p)^2+(7-q)^2}$ oe
	M1	For combining their two equations in p and q to obtain a 3TQ
	M1	For solving their 3TQ (provided it is a 3TQ) by any acceptable method. If the 3TQ is incorrect, there must be evidence of a method to score this mark. If they use their calculator on a correct 3TQ and obtain the correct values, score M1.
	A1	For the correct value of p or q
	A1	For the correct values of both p and q
(d)	M1	For any correct method to find the area of triangle ACD, using their coordinates. If they use the determinant method, check their work carefully.
	A1	For the correct area of 80 [square units].

Useful sketch



Type MS · Sector problems with an unknown radius or angle

Answer 4 of 7 · Question 4 part c · 2 marks

2024 JUN · Paper 01 (regional R) · extract 11 of 15 · find part c in this mark scheme extract

Question	Scheme	Marks
10(a)	$\alpha+\beta=-\frac{k}{2}\qquad \alpha\beta=2$	B1
	$(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta,\quad \alpha^2-\beta^2=(\alpha+\beta)(\alpha-\beta)$	B1B1
	$\Rightarrow \alpha^2-\beta^2=(\alpha+\beta)\sqrt{(\alpha+\beta)^2-4\alpha\beta}$	
	$\Rightarrow \alpha^2-\beta^2=\left(-\frac{k}{2}\right)\sqrt{\left(-\frac{k}{2}\right)^2-4\times 2}=\frac{7\sqrt{17}}{4}$	M1
	$\Rightarrow \frac{k^4}{4}-8k^2-\frac{833}{4}=0\Rightarrow k^4-32k^2-833=0$ or any equivalent	M1A1
	$\Rightarrow (k^2-49)(k^2+17)=0\Rightarrow k=-7$ *	M1A1 cso [8]
(b)	Product: $\left[(\alpha-\beta)(\alpha+\beta)=\alpha^2-\beta^2=\frac{7\sqrt{17}}{4}\right]$ from (a)	B1
	Sum: $(\alpha-\beta)+(\alpha+\beta)=\frac{\sqrt{17}}{2}+\frac{7}{2}=\left(\frac{\sqrt{17}+7}{2}\right)$	M1
	$x^2-\left(\frac{\sqrt{17}+7}{2}\right)x+\frac{7\sqrt{17}}{4}=0\Rightarrow 4x^2-2\left(\sqrt{17}+7\right)x+7\sqrt{17}=0$	M1A1 [4]
		Total 12 marks

There is an alternative solution using the roots of the equation.

Roots are $\frac{-k\pm\sqrt{k^2-4\times 2\times 4}}{2\times 2}$

$$\Rightarrow\left(\frac{-k+\sqrt{k^2-32}}{4}\right)^2-\left(\frac{-k-\sqrt{k^2-32}}{4}\right)^2=\frac{7\sqrt{17}}{4}$$

$$\Rightarrow\left(-k+\sqrt{k^2-32}\right)^2-\left(-k-\sqrt{k^2-32}\right)^2=28\sqrt{17}$$

$$\Rightarrow k^2-2k\sqrt{k^2-32}+k^2-32-k^2-2k\sqrt{k^2-32}-k^2+32=28\sqrt{17}$$

$$\Rightarrow -4k\sqrt{k^2-32}=28\sqrt{17}\Rightarrow k\sqrt{k^2-32}=7\sqrt{17}$$

$$\Rightarrow k^2\left(k^2-32\right)=833\Rightarrow k^4-32k^2-833=0$$

$$\Rightarrow (k^2-49)(k^2+17)=0\Rightarrow k=-7\ *$$

Type MS · Sector problems with an unknown radius or angle

Answer 4 of 7 · Question 4 part c · 2 marks

2024 JUN · Paper 01 (regional R) · extract 12 of 15 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	B1	For the correct sum in terms of k and the correct product. Look out for these embedded in their work.
	B1	For the correct algebra on $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta$ Substitution not required for this mark, although it can be implied from correct substitution without the actual algebra seen.
	B1	For the correct algebra on $\alpha^2-\beta^2$ Award also for $(\alpha+\beta)(\alpha-\beta)=\frac{7\sqrt{17}}{4}$ Substitution not required for this mark, although it can be implied from correct substitution without the actual algebra seen.
	M1	For substituting in the values of their sum and product into their $\alpha^2-\beta^2$ This can be implied from sight of $\left(-\frac{k}{2}\right)\sqrt{\left(-\frac{k}{2}\right)^2-4\times2}=\frac{7\sqrt{17}}{4}$
	M1	For forming a 3TQ in terms of k^2 in any form
	A1	For the correct 3TQ Accept it in any form as long as it is 3 terms. For example, accept $\frac{k^4}{16}-2k^2-\frac{833}{16}=0$ etc
	M1	For any acceptable attempt seen to solve their 3TQ [see General Guidance] If there is no working, just $k=-7$ following a 3TQ then award M0 However, accept evidence of $k^2=49$ $\left[\text{with } k^2=-17\right]$
	A1 cso	For the correct value of k * **Note: This is a given value**
(b)	B1	For a value for the product using their results from (a) or just writes it down.
	M1	For a correct method to find the value of the sum using their values for $(\alpha-\beta)$ and $(\alpha+\beta)$
	M1	For forming a 3TQ with their sum and product. = 0 is not required for the award of this mark
	A1	For a correct equation simplified or unsimplified. For example, accept, $x^2-\left(\frac{\sqrt{17}+7}{2}\right)x+\frac{7\sqrt{17}}{4}=0$ as well as $4x^2-2\left(\sqrt{17}+7\right)x+7\sqrt{17}=0$ o.e.

Type MS · Sector problems with an unknown radius or angle

Answer 4 of 7 · Question 4 part c · 2 marks

2024 JUN · Paper 01 (regional R) · extract 13 of 15 · find part c in this mark scheme extract

Question	Scheme	Marks
11(a)	$f(\theta)=(2\cos\theta-\sin\theta)(2\sin\theta+\cos\theta)$ $=4\sin\theta\cos\theta+2\cos^2\theta-2\sin^2\theta-\sin\theta\cos\theta$ $=3\sin\theta\cos\theta+2(\cos^2\theta-\sin^2\theta)\Rightarrow\frac{3}{2}\sin 2\theta,+2\cos 2\theta$ $=\frac{3}{2}\sin 2\theta+2\cos 2\theta$ *	M1 M1A1 cso [3]
(b)	$\frac{3}{2}\sin 2\theta+2\cos 2\theta+2=0\Rightarrow 3\sin 2\theta+4\cos 2\theta+4=0$ $\Rightarrow 3\sin 2\theta+4(\cos^2\theta-\sin^2\theta)+4(\sin^2\theta+\cos^2\theta)=0$ $\Rightarrow 6\sin\theta\cos\theta+8\cos^2\theta=0$ $\Rightarrow\cos\theta(6\sin\theta+8\cos\theta)=0$ $\Rightarrow\cos\theta=0\Rightarrow\theta=\frac{\pi}{2}$ * $\left[\Rightarrow\tan\theta=-\frac{4}{3}\right]$	M1M1 M1 M1 A1 cso [5]
(c)	$\text{Area}=\int_0^{\frac{\pi}{2}}\left(\frac{3}{2}\sin 2\theta+2\cos 2\theta+2\right)d\theta=\left[-\frac{3}{4}\cos 2\theta+\frac{2\sin 2\theta}{2}+2\theta\right]_0^{\frac{\pi}{2}}$ $=\left(-\frac{3}{4}\cos 2\left(\frac{\pi}{2}\right)+\frac{2\sin 2\left(\frac{\pi}{2}\right)}{2}+2\left(\frac{\pi}{2}\right)\right)-\left(-\frac{3}{4}\cos 0+\frac{2\sin 0}{2}+2\times 0\right)$ $=\left(\frac{3}{4}+\pi\right)-\left(-\frac{3}{4}\right)=\frac{3}{2}+\pi$	M1 M1 A1 [3]
		Total 11 marks

Part	Mark	Notes
(a)	M1	For multiplying out the brackets **correctly** and simplifying to $k\sin\theta\cos\theta$, $l(\cos^2\theta-\sin^2\theta)$ where k and l are integers. Condone invisible brackets if the working is clear.
	M1	For using the identity for $\cos 2\theta$ on their $2(\cos^2\theta-\sin^2\theta)$ **OR** For using the identity for $\sin 2\theta$ on their $3\sin\theta\cos\theta$
	A1 cso	For the correct identity as shown with no errors. You must check every line of working carefully – this is a given answer.
	Solutions based on RHS = LHS are acceptable. Apply the above marks – if you are not sure, please send to REVIEW	

Type MS · Sector problems with an unknown radius or angle

Answer 4 of 7 · Question 4 part c · 2 marks

2024 JUN · Paper 01 (regional R) · extract 14 of 15 · find part c in this mark scheme extract

(b)	M1	For correctly using the identity $\cos 2\theta = \cos^2\theta - \sin^2\theta$
	M1	For correctly using the identity $\sin^2\theta + \cos^2\theta = 1$
	M1	For correctly using the identity $\sin 2\theta = 2\sin\theta\cos\theta$
	M1	For factorising the resulting expression. You must see this step
	A1 cso	For $\cos\theta = 0 \Rightarrow \theta = \frac{\pi}{2}$
	ALT	
	M1	For correctly using $\cos 2\theta = \cos^2\theta - \sin^2\theta$ $\frac{3}{2}\sin 2\theta + 2\cos 2\theta + 2 = \frac{3}{2}\sin 2\theta + 2\left(\cos^2\theta - \sin^2\theta\right) = 0$
	M1	For correctly using the identity $\sin^2\theta + \cos^2\theta = 1$ $\frac{3}{2}\sin 2\theta + 2\cos 2\theta + 2 = \frac{3}{2}\sin 2\theta + 2\left(\cos^2\theta - 1 + \cos^2\theta\right) + 2 = 0$ $\Rightarrow \left[\frac{3}{2}\sin 2\theta + 4\cos^2\theta = 0\right]$
	M1	For correctly using the identity $\sin 2\theta = 2\sin\theta\cos\theta$ $\frac{3}{2}\sin 2\theta + 4\cos^2\theta = 3\sin\theta\cos\theta + 4\cos^2\theta = 0$
	M1	For factorising the resulting expression. You must see this step $3\sin\theta\cos\theta + 4\cos^2\theta = \cos\theta\left(3\sin\theta + 4\cos\theta\right) = 0$
	A1	$\cos\theta = 0 \Rightarrow \theta = \frac{\pi}{2}$ *
(c)	M1	For integrating the given expression. Minimally acceptable integration as shown below: $\frac{3}{2}\sin 2\theta \Rightarrow \pm\frac{3}{4}\cos 2\theta$ $2\theta\cos 2\theta \Rightarrow \pm\left(\frac{2}{2}\right)\sin 2\theta$ $2 \Rightarrow 2\theta$
	M1	For substituting **BOTH** of the given limits the correct way around into **their changed expression** and subtracting.. There must be a minimum of two terms to substitute limits into. This must be explicitly seen
	A1	For the correct exact area as shown. [Approximate area = 4.641….]

Type MS · Sector problems with an unknown radius or angle

Answer 4 of 7 · Question 4 part c · 2 marks

2024 JUN · Paper 01 (regional R) · extract 15 of 15 · find part c in this mark scheme extract

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1. Factorisation:

$(x^2+bx+c)=(x+p)(x+q)$, where $|pq|=|c|$ leading to $x=....$

$(ax^2+bx+c)=(mx+p)(nx+q)$ where $|pq|=|c|$ and $|mn|=|a|$ leading to $x=....$

2. Formula:

Attempt to use the **correct** formula (shown explicitly or implied by working) with values for *a*, *b* and c, leading to $x=....$

3. Completing the square:

$x^2+bx+c=0$: $\left(x\pm\frac{b}{2}\right)^2\pm q\pm c=0, \quad q\neq 0$ leading to $x=....$

Method marks for differentiation and integration

1. Differentiation

Power of at least one term decreased by 1. $\left(x^n\rightarrow x^{n-1}\right)$

2. Integration:

Power of at least one term increased by 1. $\left(x^n\rightarrow x^{n+1}\right)$

Use of a formula

Generally, the method mark is gained by **either** quoting a correct formula and attempting to use it, even if there are mistakes in the substitution of values

or, where the formula is not quoted, the method mark can be gained by implication from the substitution of correct values and then proceeding to a solution.

Answers without working

The rubric states "Without sufficient working, correct answers may be awarded no marks".

General policy is that if it could be done "in your head" detailed working would not be required. (Mark schemes may override this e.g. in a case of "prove or show...."

Type MS · Sector problems with an unknown radius or angle

Answer 5 of 7 · Question 1 part whole · 3 marks

2025 NOV · Paper 02 · extract 2 of 3 · find part whole in this mark scheme extract

Exact answers
When a question demands an exact answer, all the working must also be exact. Once a candidate loses exactness by resorting to decimals the exactness cannot be regained.

Rounding answers (where accuracy is specified in the question)
Penalise only once per question for failing to round as instructed – i.e. giving more digits in the answers. Answers with fewer digits are automatically incorrect, but the is rule may allow the mark to be awarded before the final answer is given.

Type MS · Sector problems with an unknown radius or angle

Answer 5 of 7 · Question 1 part whole · 3 marks

2025 NOV · Paper 02 · extract 3 of 3 · find part whole in this mark scheme extract

Question	Scheme	Marks
1	Let $\angle AOB = \theta$	
	$[49 = 14 + 14 + 14\theta \Rightarrow]$	B1
	$[\theta =] 1.5$ oe	
	$[A =] \frac{1}{2} \times 14^2 \times "1.5"$ oe $= 147 [\text{cm}^2]$	M1A1 [3]
	In degrees:	
	$\left[2\pi \times 14 \times \frac{\theta}{360} = (49 - 28) \right]$	
	$[\theta =] \frac{270}{\pi}$ $(85.9436...^\circ)$ oe	B1
	$[A =] \pi \times 14^2 \times \frac{"\frac{270}{\pi}"}{360}$ oe $= 147 [\text{cm}^2]$	M1A1 [3]
	$[l = r\theta = 49 - 14 - 14 =] 21$	B1
	$\left[A = \frac{1}{2} r(r\theta) = \right] \frac{1}{2} \times 14 \times 21$ oe $= 147 [\text{cm}^2]$	M1A1 [3]
		Total 3 marks

Type MS · Sector problems with an unknown radius or angle

Answer 6 of 7 · Question 3 part a · 2 marks

2025 JUN · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

Number		
3 (a)	$2\times7\times\theta=\frac{28\pi}{9}$ oe or $2\times\pi\times7\times\frac{angle}{360}=\frac{14\pi}{9}$ oe **and** attempts to change back to radians $\theta=\frac{2\pi}{9}$	M1 A1 [2]
(b)	Area of sector $A=\frac{1}{2}\times7^2\times'\frac{2\pi}{9}'$ Area of triangle $A=\frac{1}{2}\times7^2\times\sin'\frac{2\pi}{9}'$ Area of segment $[A=][2\times]\left(\frac{1}{2}\times7^2\times'\frac{2\pi}{9}'-\frac{1}{2}\times7^2\times\sin'\frac{2\pi}{9}'\right)=2.71$ (awrt)	M1 M1 ddM1A1 [4]
		Total marks 6

Part	Mark	Additional Guidance
(a)	M1	For forming a correct equation using arc length and $r=7$ If attempted by degrees, must see attempts to change degrees to radians
	A1	For $\theta=\frac{2\pi}{9}$ oee (or exact equivalent)
(b)	M1	For a correct method finding area of the sector using **their** θ $A=\frac{1}{2}\times7^2\times'\frac{2\pi}{9}'$ or $A=\pi\times7^2\times\frac{'40'}{360}$ $\left(\text{Note}:\frac{2\pi}{9}=40°\right)$
	M1	For a fully correct method finding the area of a triangle using their θ, can be implied by the correct area of triangle 15.7 awrt
	ddM1	For finding the difference between the area of the sector and the area of the triangle. The subtraction must be shown, or it could be implied by correct final answer. Dependent on **both** previous M marks
	A1	For 2.71 (awrt) ignore unit

Type MS · Sector problems with an unknown radius or angle

Answer 6 of 7 · Question 3 part a · 2 marks

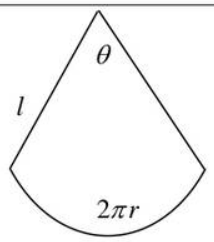
2025 JUN · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Sector problems with an unknown radius or angle

Answer 7 of 7 · Question 10 part a · 4 marks

2020 NOV · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

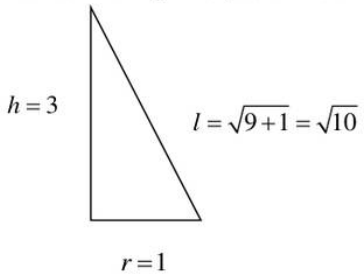
Number			
10(a)	Circumference of base $= 2\pi r$ 		B1
	$l\theta = 2\pi r \Rightarrow \theta = \dfrac{2\pi r}{l}$		B1
	$A = \dfrac{1}{2}l^2\theta = \dfrac{1}{2}l^2\dfrac{2\pi r}{l} = \pi rl$		M1A1 (4)
(b)	$A = \pi rl$		
	$l = r\sqrt{10} \Rightarrow A = \pi r^2\sqrt{10}$		B1
	$\dfrac{dA}{dr} = 2\pi r\sqrt{10} \Rightarrow k = 2\sqrt{10}$		M1A1 (3)
(c)	$\dfrac{dV}{dt} = 1.5\ \left(\text{cm}^3/\text{s}\right)$		B1
	$\dfrac{dA}{dt} = \dfrac{dA}{dr} \times \dfrac{dr}{dV} \times \dfrac{dV}{dt}$		M1
	$V = \dfrac{1}{3}\pi r^2 h = \dfrac{1}{3}\pi r^2 \times 3r = \pi r^3$		B1
	$\dfrac{dV}{dr} = 3\pi r^2$		B1ft
	$\therefore \dfrac{dA}{dt} = 2\pi \times 8\sqrt{10} \times 1.5 \times \dfrac{1}{3 \times 64\pi} = 0.3952... = 0.395\ \text{cm}^2/\text{s}$		A1 (5)
			[12]

Part	Mark	Notes	
(a)	B1	For the circ of the base $L = 2\pi r$	
	B1	$R = l$ and $l = r\theta$ Therefore $l\theta = 2\pi r \Rightarrow \theta = \dfrac{2\pi r}{l}$	$R = l$
	M1	$A = \dfrac{1}{2}l^2\theta$ and substituting their expression for θ to give $A = \dfrac{1}{2}l^2\theta = \dfrac{1}{2}l^2\dfrac{2\pi r}{l}$	Uses the formula $A = \dfrac{1}{2}RL$ $A = \dfrac{1}{2} \times l \times 2\pi r \Rightarrow \left(A = \pi rl\right)$
	A1	For the required expression for A, $A = \pi rl$ with no errors.	

Type MS · Sector problems with an unknown radius or angle

Answer 7 of 7 · Question 10 part a · 4 marks

2020 NOV · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

(b)	B1	For finding that the slant height is $\sqrt{10}$ times the radius of the cone  So $l=r\sqrt{10}$
	M1	Substitutes $l=r\sqrt{10}$ into the given expression $A=\pi rl$ and differentiates their resulting expression to find $\frac{dA}{dr}$ $A=\pi r^2\sqrt{10}$ so $\frac{dA}{dr}=2\pi r\sqrt{10}$
	A1	Therefore $k=2\sqrt{10}$
(c)	B1	States $\frac{dV}{dt}=1.5\ \left(\text{cm}^3/\text{s}\right)$ Award if it seen explicitly in (b) **and used** in (c)
	M1	States (or uses) a correct chain rule $\frac{dA}{dt}=\frac{dA}{dr}\times\frac{dr}{dV}\times\frac{dV}{dt}$
	B1	For finding the volume of a cone in terms of r only $V=\frac{1}{3}\pi r^2h=\frac{1}{3}\pi r^2\times 3r=\pi r^3$
	B1ft	Differentiates their expression for the volume of a cone provided it is in terms of V and r only. $\frac{dV}{dr}=3\pi r^2$
	A1	For combining all required terms into their chain rule and evaluating to 3 significant figures, $\frac{dA}{dt}=2\pi\times 8\sqrt{10}\times 1.5\times\frac{1}{3\times 64\pi}=0.3952...=0.395\ \text{cm}^2/\text{s}$
	ALT – in terms of h	
	B1	States $\frac{dV}{dt}=1.5\ \left(\text{cm}^3/\text{s}\right)$ Award if it seen explicitly in (b) **and used** in (c)
	M1	States (or uses) a correct chain rule $\frac{dA}{dt}=\frac{dA}{dh}\times\frac{dh}{dt}$ and $\frac{dV}{dt}=\frac{dV}{dh}\times\frac{dh}{dt}$
	B1	For finding the area and volume of a cone in terms of h $r=\frac{h}{3}$, $A=\pi r^2\sqrt{10}\Rightarrow A=\frac{\sqrt{10}}{9}\pi h^2$ and $V=\frac{1}{3}\pi r^2h\Rightarrow V=\frac{1}{27}\pi h^3$
	B1ft	Differentiates their expressions for the area and volume of a cone provided they are both in terms of h only. $\frac{dA}{dh}=\frac{2\sqrt{10}}{9}\pi h$ and $\frac{dV}{dh}=\frac{3}{27}\pi h^2$
	A1	Combines the required terms into their chain rules and evaluating to 3 significant figures $\frac{dA}{dt}=0.395\ 102$

Type MS · Sector problems with an unknown radius or angle

Answer 7 of 7 · Question 10 part a · 4 marks

2020 NOV · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

Question Number	Scheme	Marks
	$\frac{1}{2}$	